

Materials and Methods

Initially for Writing a better Comparison of unified architecture of detection and embedding for livestock ATRW dataset is chosen. The ATRW Dataset is a large-scale dataset for animal re-identification, which contains images of various animal species captured in the wild.

Dataset

The Amur Tiger Re-identification in the Wild (ATRW) dataset was used to evaluate the unified detection-and-embedding architecture. The dataset is organized into two subsets: a *detection* subset, providing bounding-box annotations for localizing tigers in camera-trap imagery, and a *re-identification (re-ID)* subset, providing identity and keypoint annotations for distinguishing individual tigers by their unique stripe patterns.

The detection subset contains 4,413 images in total, split into a trainval partition of 2,762 images (further divided into 2,485 images for training and 277 images for validation) and a held-out test partition of 1,651 unannotated images used for detection evaluation. Bounding-box annotations for the 2,762 trainval images are provided in PASCAL VOC XML format.

The re-ID subset contains 5,156 images. The training pool comprises 3,392 images, of which 1,887 images are identity-labeled, covering 107 unique tiger identities, while the remaining 1,505 images are unlabeled and used as an additional gallery pool for the plain and cross-camera evaluation protocols. The test partition consists of 1,764 unlabeled query images used for identity-retrieval evaluation. Body-part keypoint annotations, following the MSCOCO keypoint format, are provided for all labeled training images and all test images.

Table 1: Composition of the ATRW dataset used in this study

Task	Split	Images	Notes
Detection	Train	2,485	Subset of trainval; bounding-box annotations (PASCAL VOC XML)
	Validation	277	Subset of trainval; bounding-box annotations (PASCAL VOC XML)
	Test	1,651	Unannotated; used for detection evaluation only
Re-identification	Train (labeled)	1,887	107 unique tiger identities; MSCOCO-format keypoints
	Train (gallery pool)	1,505	Unlabeled; additional images for plain/cross-camera evaluation
	Test (query)	1,764	Unlabeled query images; MSCOCO-format keypoints

On disk, the detection subset totals approximately 3.6 GB (2,762 trainval and 1,651 test images), while the re-ID subset totals approximately 239 MB (3,392 train and 1,764 test images), giving a combined dataset size of roughly 3.84 GB used for training and benchmarking the unified detection-and-embedding architecture.

Level 2 heading

$$e^{\pi i} + 1 = 0 \tag{1}$$

Level 3 heading

Results

Type or paste your results here as prescribed by the journal's instructions for authors. Avoid redundant data that from the tables or illustrations. Type or paste your results here as prescribed by the journal's instructions for authors. Type or paste your results here as prescribed by the journal's instructions for authors (Table 2). Type or paste your results here as prescribed by the journal's instructions for authors. Type or paste your results here as prescribed by the journal's instructions for authors. Type or paste your results here as prescribed by the journal's instructions for authors.

Table 2: My caption

Word No.	Population	Area (Acre)	Estimated waste generation		
			Kg/capita/day	Kg/day	Percent
1	18,250	747	0.27	4,928	31
2	9,500	38	0.29	2,755	17
3	12,200	181	0.15	1,830	12
Total	103,903	3,316	0.15	15,700	100

Here is a footnote.

Type or paste your results here as prescribed by the journal's instructions for authors. Type or paste your results here as prescribed by the journal's instructions for authors. Type or paste your results here as prescribed by the journal's instructions for authors (Figure 1). Type or paste your results here as prescribed by the journal's instructions for authors.



Figure 1: Type your title here. Obtain permission and include the acknowledgement required by the copyright holder if a figure is being reproduced from another source.

Discussions

Type or paste your discussion here as prescribed by the journal's instructions for authors. Type or paste your discussion here as prescribed by the journal's instructions for authors. Type or paste your discussion here as prescribed by the journal's instructions for authors. Type or paste your discussion here as prescribed by the journal's instructions for authors. Type or paste your discussion here as prescribed by the journal's instructions for authors. Type or paste your discussion here as prescribed by the journal's instructions for authors. Type or paste your discussion here as prescribed by the journal's instructions for authors. Type or paste your discussion here as prescribed by the journal's instructions for authors. Type or paste your discussion here as prescribed by the journal's instructions for authors. Type or paste your discussion here as prescribed by the journal's instructions for authors.

Conclusion

In this part, authors should conclude the significance of the study, emphasize its value and state expectation on future studies that may need to be carried out. In details, it may include summary of key findings, strengths and limitations of the study, controversies raised by this study, and future research directions, etc.

Acknowledgement

Type or paste your Acknowledgement here as prescribed by the journal's instructions for authors. This section is not added if the author does not have anyone to acknowledge.

References

Please follow the **APA 7th edition** style for your references that is available at <https://guides.unitec.ac.nz/apareferencing>. The author should follow APA 7th edition reference style both in inside text and reference. Here is a sample to cite references (Shome, 2022).

References

Shome, A. (2022). Khulna university studies journal template. *Khulna University Studies*, 1.0.